

Somjit Nath

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Education

- 2022 – present **Doctor of Philosophy (PhD) in Engineering**, MILA & École de technologie supérieure (ETS), Montreal, Canada.
Supervisor: Prof. Samira Ebrahimi Kahou
- 2017 – 2019 **Master of Science (Thesis) in Computing Science**, University of Alberta, Edmonton, Canada.
CGPA: 4.00/4.00
Supervisor: Prof. Martha White
- 2013 – 2017 **Bachelor of Engineering in Electrical Engineering**, Jadavpur University, Kolkata, India.
First Class Honours, CGPA: 9.08/10.00, 85.52%
- 2013 **Higher Secondary Examination**, South Point High School, Kolkata, India.
Science (Mathematics, Physics, Chemistry); Elective - Statistics.
First Division, 88.4%
- 2011 **Madhyamik**, South Point High School, Kolkata, India.
Elective - Mechanics.
First Division, 90%

Experience

- 2019 – 2021 **Researcher**, Planning & Control, Tata Consultancy Services Research & Innovation, Mumbai, India.
Supervisor: Dr. Harshad Khadilkar

Publications

- Feb 2022 **Follow your Nose: Using General Value Functions for Directed Exploration in Reinforcement Learning**, *Reinforcement Learning in Games Workshop, Association for the Advancement of Artificial Intelligence(AAAI), 2022*, Somjit Nath, Omkar Shelke, Durgesh Kalwar, Hardik Meisheri, Harshad Khadilkar.
- Nov 2021 **Revisiting State Augmentation methods for Reinforcement Learning with Stochastic Delays**, *Conference on Information and Knowledge Management (CIKM) 2021*, Somjit Nath, Mayank Baranwal and Harshad Khadilkar.

- May 2021 **Scalable Multi-Product Inventory Control with Lead Time Constraints using Reinforcement Learning**, *Neural Computing and Applications*, Hardik Meisheri, Nazneen N Sultana, Mayank Baranwal, Vinita Baniwal, Somjit Nath, Satyam Verma, Balaraman Ravindran, Harshad Khadilkar.
- May 2021 **SIBRE: Self Improvement Based REwards for Adaptive Feedback in Reinforcement Learning**, *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021, Somjit Nath, Richa Verma, Abhik Ray, Harshad Khadilkar.
- April 2020 **Training Recurrent Neural Networks Online by Learning Explicit State Variables**, *International Conference on Learning Representations (ICLR)*, 2020, Somjit Nath, Vincent Liu, Alan Chan, Xin Li, Adam White, Martha White.
- May 2019 **Two-Timescale Networks for Nonlinear Value Function Approximation**, *International Conference on Learning Representations (ICLR)*, 2019, Wesley Chung, Somjit Nath, Ajin Joseph, Martha White.
- Dec 2018 **A Fixed-Point Formulation for Recurrent Neural Networks**, *Continual Learning Workshop, Neural Information Processing Systems(NeurlPS)*, 2018, Somjit Nath, Taher Jafferjee and Martha White.
- Dec 2018 **Rejection Sampling for Off-Policy Learning**, *Continual Learning Workshop, Neural Information Processing Systems(NeurlPS)*, 2018, Wesley Chung, Sina Ghiassian, Somjit Nath and Martha White.
- April 2018 **Smartphone Camera Based Analysis of ELISA using Artificial Neural Network**, *IET Computer Vision*, Somjit Nath, Subhannita Sarcar, Biswendu Chatterjee, Nabendu Chatterjee, Rhishita Chourashi.
- Dec 2016 **Arduino Based Door Unlocking System with Real Time Control**, *IEEE International Conference on Contemporary Computing and Informatics (IC3I)*, Somjit Nath, Paramita Banerjee, Rathindra Nath Biswas, Swarup Kumar Mitra and Mrinal Kanti Naskar.

Teaching Experience

- Jan - April 2018 **Graduate Teaching Assistant**, *Department of Computing Science, University of Alberta, Edmonton, Canada.*
CMPUT 275, Introduction to Tangible Computing-II
- Sept - Dec 2017 **Graduate Teaching Assistant**, *Department of Computing Science, University of Alberta, Edmonton, Canada.*
CMPUT 274, Introduction to Tangible Computing-I

Internships

June 2016 **Vocational Trainee**, *Steel Authority of India IISCO*, Burnpur, India.
May – Jul 2015 **Research Intern**, *Indian Statistical Institute*, Kolkata, India.
Computer Vision and Pattern Recognition Department

Awards & Achievements

Feb 2022 **TCS Citation Award**, *Tata Consultancy Services*.
Oct 2021 **TCS Citation Award**, *Tata Consultancy Services*.
May 2017 **Scholarship for Academic Excellence**, *State Electrical Engineers' Association*, Kolkata, India.
Feb 2016 **2nd in Autonomous Robotics event "Sherlock"**, *KSHITIJ*, the technology fair (largest in Asia) held at *Indian Institute of Technology(IIT)*, Kharagpur, India.
2015 **Summer Fellow**, *Indian Academy of Sciences*, Bengaluru, India.
Feb 2014 **2nd in Autonomous Robotics event**, *SRIJAN*, the technology fair held at *Jadavpur University*, Kolkata, India.

Technical skills

Language C,C++,Python,HTML,PHP,Javascript,SQL,VHDL
Tools Tensorflow, PyTorch, MATLAB, Octave, OpenCV, Arduino, Processing
Operating System Windows, UNIX, MacOSX

Extra-curricular Activities

2018 Love to play Table Tennis and participated in tournaments – Edmonton Chinatown Open 2018 & CUSTTA Open 2018, Calgary
Mar 2016 Member of the organizing committee of the hackathon "SPARKHACK" and the circuit-building competition "CIRCUISTIC" held at the annual department technology fair "CONVOLUTION '16", Department of Electrical Engineering, Jadavpur University, Kolkata, India
Mar 2016 Mentored a group of five school students for a robotics competition in "CONVOLUTION' 16", Department of Electrical Engineering, Jadavpur University, Kolkata, India